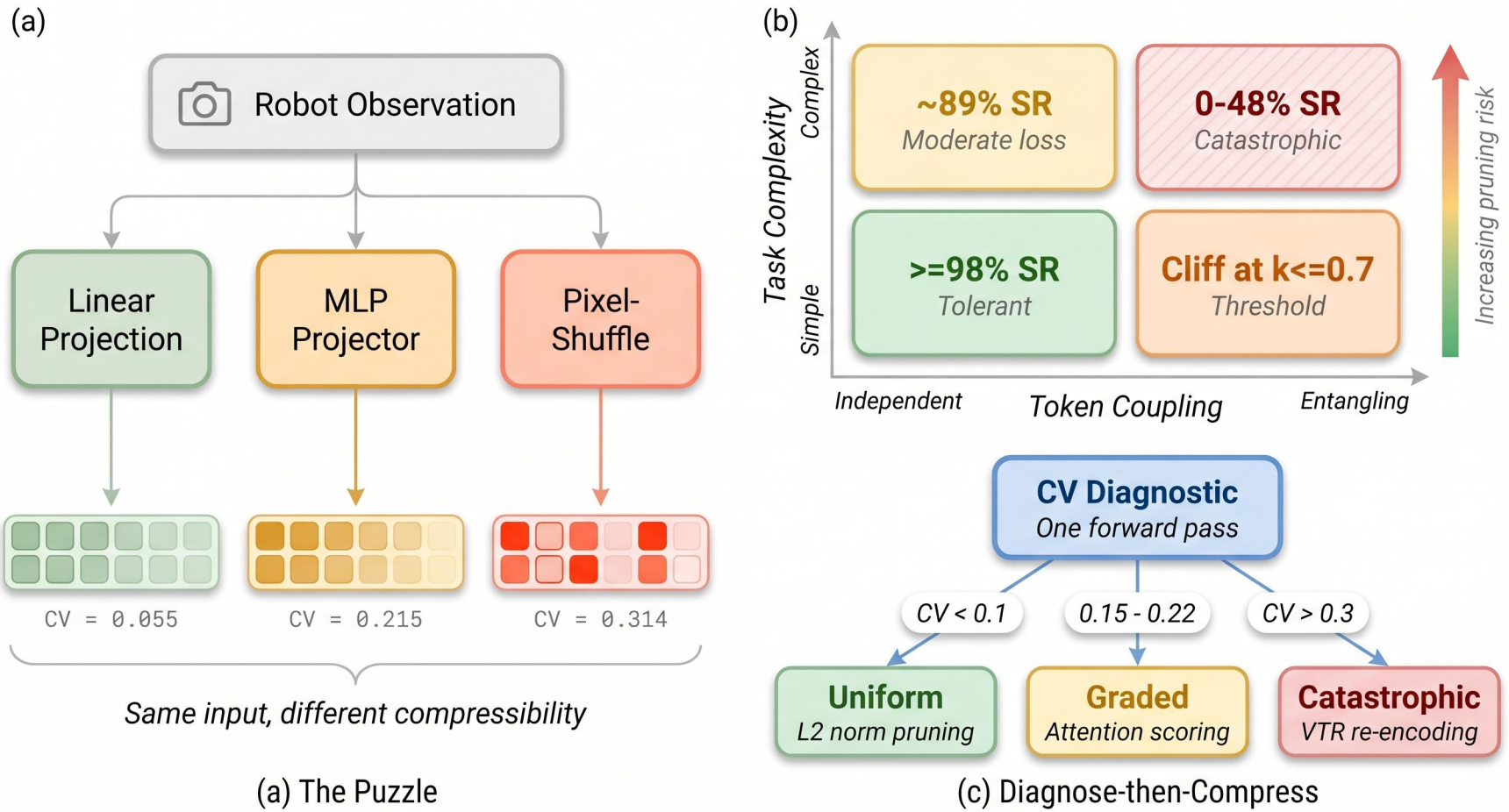

(a)
Robot Observation
Linear Projection
MLP Projector
Pixel-Shuffle
CV = 0.055
CV = 0.215
CV = 0.314
Same input, different compressibility
(a) The Puzzle
(b)
Task Complexity
Complex
Simple
~89% SR
Moderate loss
0-48% SR
Catastrophic
>=98% SR
Tolerant
Cliff at k<=0.7
Threshold
Increasing pruning risk
Independent
Token Coupling
Entangling
CV Diagnostic
One forward pass
CV < 0.1
0.15 - 0.22
CV > 0.3
Uniform
L2 norm pruning
Graded
Attention scoring
Catastrophic
VTR re-encoding
(c) Diagnose-then-Compress